WebRouteMap: Which Networks Carry Your Web Traffic?

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Abstract

Web access depends on intermediary networks that are difficult for users and content providers to see. A path view could help users inspect transit dependencies and providers compare access to serving locations, but direct measurements require vantage points in the relevant user networks. We present WebRouteMap, a tool that estimates the autonomous-system (AS) paths behind Web access without measurements from every target user network. Given a user network and a selected website endpoint, it combines public routing observations with interconnection metadata to rank possible exits, extend branches with sparse continuations, and score complete paths using destination-specific propagation evidence. A shared selected path library supports training and candidate generation; an existing ranker then answers individual path queries. We evaluate a controlled BGP source-withholding experiment, whose destinations represent potential Web-service locations, and independent traceroutes to domain-derived endpoints. The three-snapshot BGP benchmark obtains approximately 80% AS-set Jaccard, using ten source ASes per date. The single-source Web experiment obtains 69.96% over 1,890 pairs, compared with 69.19% for KnownPath. The evaluations quantify network-path agreement for Web access.

CCS Concepts

• **Networks** → **Network measurement**.

Keywords

Web access, network-path visibility, public routing evidence, AS-level routing

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1 Introduction

Which networks carry a user's traffic when visiting websites? Domain names identify services but reveal little about the intermediary networks connecting users to their serving endpoints. A user reading news, shopping, and streaming video may depend on a common transit network despite visiting different services (Figure 1). At the autonomous-system (AS) level, visibility into these paths could help users inspect their exposure to intermediary networks and help content providers compare how user networks reach candidate serving locations. This view complements studies of hosting, DNS, and certificate dependencies [7, 11] by examining the networks connecting users to Web services.

Obtaining this visibility across user networks is difficult. Traceroute observes paths from deployed probes, so coverage depends on suitable vantage points [9, 28]. Public BGP collectors add routing observations from participating networks, but their coverage

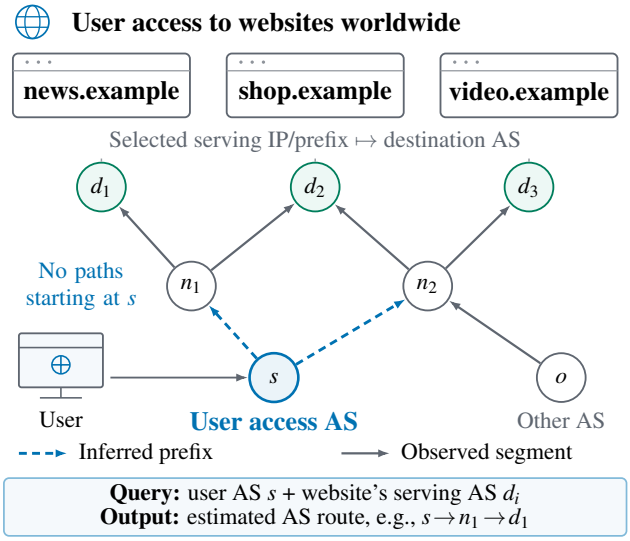


Figure 1: Network paths behind Web access. For each externally supplied serving AS d_i, WebRouteMap joins an estimated prefix with an observed segment to estimate the user's route. Dashed and solid arrows distinguish the two sources of evidence. The diagram shows AS connectivity, not geography.

is incomplete [8, 29, 32]. Path inference extends these observations. Graph-based methods can generate routes without source-originated measurements, yet simplified policy rules cannot fully capture source-specific routing behavior [10, 25]. Stitching methods reuse observed segments, but composition alone cannot construct a route when suitable source-connected segments are absent [16]. Hybrid methods improve coverage, while uneven observations remain a challenge across source networks [10].

The shortage of source observations creates a cold-start problem for learning-based inference. For example, GMPI trains a source-specific model using observed paths rooted at the source AS [19]; few or no such paths leave insufficient supervision for learning its routing behavior [18]. PA-GMPI introduces adaptive training and warm-start mechanisms [17]. We ask: *Can we make the network paths behind Web access visible to users and content providers, without requiring path measurements from every user network?* To address this cold start, we use public interconnection information to connect the source to route segments observed from other networks. This path augmentation provides plausible source-rooted training paths for learning routes from user networks to website endpoints.

Constructing these training paths presents three challenges. **First, how should likely forwarding branches be identified and explored?** Well-observed neighbors can supply many reusable paths, whereas poorly observed neighbors require further exploration. When a source has few neighbors, important routing choices

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may lie beyond its first hop. Neighbor selection and recursive exploration must therefore determine which branches to follow and where usable observations become available. **Second, how should paths be generated and selected?** Graph search or stitching can produce paths the source would not use. Construction must be coupled with reliability assessment and selection that preserves important neighbors. **Third, how can public evidence inform these decisions without AS configurations or direct source measurements?** BGP paths, AS relationships, and interconnection metadata describe different aspects of routing and have uneven coverage. Their information requires representations of ASes, forwarding transitions, and complete paths that account for source and destination context while preserving missing or uncertain evidence.

We present **WebRouteMap**, a Web-path visibility framework that estimates routes from user networks to selected website endpoints using public routing observations and interconnection metadata.

To address the first challenge, we combine learned neighbor ranking with evidence-guided recursive exploration. A source-level ranker prioritizes neighbors through which the source is likely to forward traffic. For each selected neighbor, we count observed continuations that can form structurally admissible training paths. Sufficient evidence allows direct reuse; otherwise, a local ranker identifies supported next hops for bounded prefix expansion. We retain prefixes with usable continuations alongside their extensions, preserving sparse evidence while exploring routing choices beyond the source's immediate neighbors.

To address the second challenge, we combine observed-path stitching with learned complete-path scoring and neighbor-aware selection. Observed BGP paths encode routing choices made by real networks, so reusing intact segments reduces how much of a training path must be inferred. We connect the source to segments rooted at selected neighbors or recursively reached ASes, preserving their direction and order. A learned scorer estimates observation support for the complete candidate. Selection reserves available samples through important neighbors before filling the remaining budget with high-scoring alternatives, keeping important forwarding branches represented in the training library.

To address the third challenge, we build a multilevel representation of public routing and interconnection evidence. AS features describe network roles, connectivity, and PeeringDB co-location; transition features capture directed commercial relationships and local routing support; path features summarize prefix reliability and observed segment support. Destination-specific evidence measures agreement among other source networks under comparable incoming commercial roles. Per-source normalization and equal source weighting limit dominance by prolific observers, while explicit missingness and conflict indicators preserve uncertainty. These representations support neighbor ranking, recursive exploration, and complete-path assessment without private AS configurations.

Our implementation uses GMPI for path inference, with the selected library supplying training paths and evidence for candidate generation in user-to-website queries such as those in Figure 1.

We make three contributions:

- We present WebRouteMap, a framework that estimates AS-level paths from user networks to selected website endpoints without requiring path measurements from the target user network.
- We develop an evidence-guided path-augmentation method for ASes without source-originated measurements. Using multilevel public evidence, it combines learned neighbor selection, recursive exploration, and observed-segment stitching with complete-path scoring and sample safeguards for important neighbors.
- We validate WebRouteMap in two complementary settings: a controlled BGP source-withholding benchmark across three snapshots and an independent traceroute experiment toward domain-derived endpoints. Stage-wise evaluation measures neighbor coverage and augmented-path observation support, alongside final AS-set Jaccard.

2 Background and Related Work

2.1 Web Access and Interdomain Routes

A user's route to a website depends on its access network and serving endpoint. DNS steering can select different server addresses, while anycast can reach different serving instances at the same address [13, 14, 30]. Mapping the endpoint prefix to an AS defines an interdomain query, although prefixes within that AS may follow different routes. This view complements Web-dependency research [7, 11] by examining the networks connecting users to selected endpoints.

BGP distinguishes learning a route from a neighbor, selecting it under local policy, and exporting it to other neighbors [26]. Route Views and RIPE RIS expose paths advertised by participating networks [29, 32]. CAIDA relationships, ASRank, and PeeringDB add commercial roles, network attributes, and shared IXP or facility presence [1, 5, 24]. These sources constrain possible routes, but do not disclose all private preferences. Co-location does not establish a connection, and valley-free compatibility does not establish that a network uses a particular route [6].

2.2 Direct Path Measurement

Traceroute varies the IP time-to-live to collect replies from forwarding hops [9]. Its output describes an observation from a particular source toward a target at the measurement time. Classic traceroute can combine replies from different paths under load balancing; Paris traceroute controls flow identifiers to reduce per-flow artifacts [2]. It does not eliminate every form of load balancing or measurement uncertainty. Scamper supports parallel, rate-controlled probing, while RIPE Atlas provides distributed probes for user-defined measurements [20, 28].

These tools support large-scale measurement, but directly observing paths from a user AS requires an available vantage point there [28]. Repeated probing consumes measurement resources. Unresponsive hops and filtering can leave gaps [27], and converting IP replies into AS sequences introduces mapping uncertainty [23]. BGP paths provide control-plane evidence and need not match a traceroute's data-plane sequence. Direct measurement and inference are complementary: measurements supply observations, while

inference estimates paths from user networks without a suitable source probe.

2.3 AS Path Inference

AS-path inference estimates unmeasured routes from available evidence [18]. **KnownPath** exploits AS paths already present in BGP tables to infer routes from a source AS to a destination prefix [25]. **ProbInfer** represents observed path segments in a multigraph and combines segment composition with probability-based candidate evaluation, using public passive measurement data [16].

GMPI combines heuristic candidate-path generation with a neural evaluation model. It integrates topology and AS attributes with related observed paths through path representations and dual attention [19]. **PA-GMPI** extends this approach with personalized path generation and adaptive model training using information specific to individual ASes. It introduces a graph-kernel-based training warm starter and an AS representation learner based on observed walks [17]. These works address candidate generation, evidence integration, and source-specific inference; we reuse the existing GMPI-based downstream ranking stage.

PathRadar distinguishes core and shell ASes according to routing complexity and observation adequacy, combining policy-based and stitching-based inference. Its progressive learning process propagates and evaluates candidate routes; it also addresses prefix-specific paths, multiple paths per AS pair, and selected non-valley-free path patterns [10].

Other systems provide complementary perspectives: iPlane composes observed segments to predict paths and their properties [21, 22]; PredictRoute combines historical data- and control-plane evidence with probabilistic path prediction [31]; and RouteInfer models diversity and commonality in ISP routing behavior [33]. Our focus is source-specific evidence construction for Web-path queries when paths originating from the user network are unavailable. We combine learned neighbor selection, expansion of evidence-poor branches, and complete-path support scoring to produce one selected library for both training and candidate generation. This construction complements the downstream inference model.

3 Problem Setting

3.1 Web-Path Queries and Scope

A query supplies a source AS s, destination AS d obtained from a selected website endpoint, and routing snapshot t. The tool returns an ordered forward AS sequence $\widehat{P}_{s,d}$ or reports that no usable candidate is available. The same snapshot allows evidence to be reused across endpoint queries. This is the tool's output; constructed paths used to answer the queries are intermediate evidence.

The AS-pair abstraction groups prefixes and endpoints within the same AS. It estimates interdomain connectivity, not routers, geography, the return path, or the forwarding path of every Web request. The BGP experiment models destination ASes as potential Web-service networks; it does not establish that every such AS hosts a website. The domain-endpoint experiment separately tests destinations obtained by resolving a popular-domain list.

Table 1: Query inputs and intermediate evidence.

Symbol	Meaning
s, d	User-network and selected endpoint AS
$\mathcal{P}, \mathcal{G}$	Public path corpus and its AS graph
$\Gamma(u), \rho(u, v)$	Graph neighbors and directed AS relationship
$\mathcal{N}_s, \mathcal{A}_{s,n}$	Admitted first hops and recursive anchors
$T_{z,j}(P), F(T)$	Directed segment rooted at z, and its frequency
$\mathcal{U}_s, \mathcal{K}_s$	Augmentation shortlist and observed source-rooted suffixes
$\mathcal{R}_s, \pi_s(n)$	Selected path library and predicted first-hop importance
$C_{s,d}, \widehat{P}_{s,d}$	Query candidates and returned route estimate

3.2 Public Evidence and Reusable Path Segments

Let $\mathcal{P}$ be the collected AS paths available at a snapshot, excluding paths that start at the target source. The source may occur internally in a path from another network. We build a graph $\mathcal{G}$ from adjacent ASes and use archived relationship and network-metadata bundles. Table 1 defines the principal symbols.

For $P = (v_0, \ldots, v_\ell) \in \mathcal{P}$ with $z = v_i$, a reusable segment is

$$T_{z,j}(P) = (v_i, \ldots, v_j), \qquad i < j \le \ell. \tag{1}$$

Segments in (1) retain their observed direction and interior AS sequence; their endpoint may be internal to the parent path. The special case $z = s$ and $j = \ell$ yields a complete source-rooted suffix when s occurs internally. We denote these original-terminal suffixes by $\mathcal{K}_s$. We never reverse an observation to manufacture a forward route.

3.3 Training Library and Inference Objective

We seek a source-specific library $\mathcal{R}_s$ that supplies training paths for learning the source's routing behavior without measurements originating from s. The library contains augmented paths and eligible observed suffixes from $\mathcal{K}_s$, retaining examples through important forwarding neighbors. The same library supports model training and local candidate generation. Given a candidate set $C_{s,d}$, the query objective is to select

$$\widehat{P}_{s,d} = \arg\max_{P \in C_{s,d}} f_{\theta_s}(P, H_{s,d}), \tag{2}$$

where $H_{s,d}$ is the available query-specific path history and f_{θ_s} is a learned source-specific path score. Library construction provides reusable supervision; query inference chooses one route for the requested endpoint. Scores express relative preference among candidates.

4 WebRouteMap Design

4.1 Framework Overview

WebRouteMap estimates the routes behind Web access using a source-specific path library $\mathcal{R}_s$ constructed from other networks' observations and public interconnection information. For each source and snapshot, learned neighbor ranking and recursive exploration identify usable branches. Stitching reuses observed segments; complete-path scoring and selection admit candidates into the library.

Figure 2 separates path augmentation (Steps 1–3) from website queries. Shared AS, transition, and path representations support

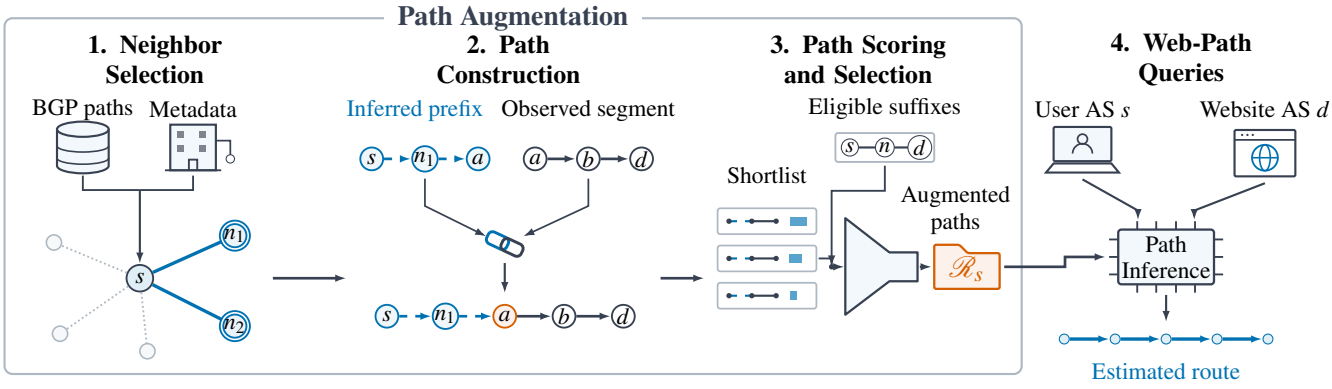


Figure 2: WebRouteMap groups the first three steps as path augmentation, producing a library for model training and Web-path queries. Dashed prefixes join solid observed segments; observed suffixes enter the shared scoring and selection stage.

Table 2: Public evidence at AS, transition, and path levels, complemented by destination context and uncertainty indicators.

Level	Representation and use
AS and AS pair	Network size and transit role; graph connectivity and overlap; shared PeeringDB locations. Used to rank possible neighbors.
Transition	Preceding, current, and next AS; directed commercial roles, local support, and prefix-phase compatibility. Used for exploration and prefix assessment.
Complete path	Ordered edge and segment support, prefix reliability, actual third AS, and continuation evidence. Used to score constructed routes.
Destination context	Donor agreement toward the same destination, with incoming-role conditioning and equal donor-source weighting; evaluated at the first hop and stitching boundary.
Uncertainty	Missing metadata, unknown or conflicting roles, absent donor coverage, and the end of a construction boundary.

the construction decisions. The selected library supplies training paths and local evidence for candidate generation, and is reused across website endpoints.

4.2 Representing Public Routing Evidence

We encode AS relationships, graph structure, network attributes, PeeringDB co-location, and observed paths as tabular features at three levels (Table 2) [1, 5, 24]. Provenance and missing or conflicting evidence remain explicit. Co-location supplies candidate information without establishing adjacency.

Destination-conditioned routing support. An observed route at a neighbor need not be available to the source: learning, selecting, and exporting a BGP route are distinct operations [26]. For a candidate $(s, n, a, \ldots, d)$, we therefore measure whether other source networks observe n followed by a toward the same destination d. Let $c_v(n, a, d)$ count distinct observed paths beginning (v, n, a) and ending at d, and let $c_v(n, d) = \sum_b c_v(n, b, d)$. With $V_{n,d} = \{v : c_v(n, d) > 0\}$, we compute

$$f(n, a, d) = \frac{|\{v \in V_{n,d} : c_v(n, a, d) > 0\}|}{|V_{n,d}|},$$
$$m(n, a, d) = \frac{1}{|V_{n,d}|} \sum_{v \in V_{n,d}} \frac{c_v(n, a, d)}{c_v(n, d)}. \quad (3)$$

Here f measures agreement across donor sources, while m also accounts for each donor's alternative next hops. Per-donor normalization and equal donor weighting limit the influence of prolific observers. We also compute these statistics using donors whose incoming commercial relationship to n matches that of the candidate's preceding AS. Only original first-hop contexts $(v, n, a, \ldots, d)$ contribute; arbitrary internal occurrences of n do not establish this export context.

At the first hop and prefix–segment boundary, two 16-feature context vectors and one boundary indicator add 33 features to 181 route, 32 prefix-reliability, and 9 actual-next-hop features: 255 in total. Absent donor coverage, unknown roles, and conflicts have explicit indicators.

4.3 Neighbor Ranking and Recursive Exploration

Learning likely forwarding neighbors. For s, we retrieve candidate neighbors from the public AS graph, remaining BGP observations, and PeeringDB co-location, prioritizing graph-supported candidates within a pool of at most 4,096 ASes. We train LightGBM [12] ranking and presence models using public and structurally extended views. The latter adds degree-normalized overlap and population-weighted co-location features. Training labels give each destination equal weight, divided across its distinct observed routes. Neighbors are ordered by this first-hop support: Top-3 neighbors receive relevance 3, ranks 4–10 receive 2, other observed neighbors receive 1, and unobserved neighbors receive 0. Presence labels indicate observed first-hop use.

Let $r_s^{\mathrm{str}}(n)$ be the structural ranking score and $p_s^{\mathrm{pub}}(n), p_s^{\mathrm{str}}(n)$ the two presence scores. We use

$$\bar{p}_s(n) = \tfrac{1}{2}\big(p_s^{\mathrm{pub}}(n) + p_s^{\mathrm{str}}(n)\big),$$
$$u_s(n) = z\big(r_s^{\mathrm{str}}(n)\big) + 0.5 \log \frac{\bar{p}_s(n)}{1-\bar{p}_s(n)}, \tag{4}$$

where z standardizes scores within the candidate pool and probabilities are clipped before taking the logit. The union of both rankers' Top-3 candidates forms a protected core. We place it first, ordering within the core and the remaining pool by u_s, then select the longest prefix containing the core with at most 32 neighbors and mean $\bar{p}_s \geq 0.7$. If no such prefix exists, we retain the core and mark its support uncertain. The output is $\mathcal{N}_s$; individual selected neighbors need not exceed 0.7.

A separate grouped-softmax head learns destination-balanced first-hop importance from paths containing at least three ASes. For sample allocation, we normalize its predicted masses over selected neighbors to obtain $\pi_s(n)$. These are allocation weights derived from observed route support, rather than estimates of traffic volume.

Exploring branches with insufficient observations. For a prefix $q = (s, n, \ldots, z)$, let $c(q)$ count distinct observed forward tails that yield structurally admissible paths when joined to q. Counting stops once τ usable tails are found. We retain every prefix with $c(q) > 0$ for construction. When $c(q) < \tau$, a local next-hop ranker conditioned on the preceding AS and commercial roles identifies supported extensions. An added edge requires observed or public connection evidence, or co-location accompanied by a finite learned support score. Extensions exclude repeated ASes and conflicts with the prefix's possible commercial phases.

We use $\tau = 300$, at most ten next hops per extension, and a maximum prefix of three ASes, yielding (s, n) and (s, n, a) prefixes. Recursion supplements usable direct prefixes; its trigger is observation availability, not source degree. Expansion can expose upstream choices for sources with few exits. A branch stops at the prefix limit or when no supported continuation is available.

4.4 Path Construction and Selection

Reusing observed routing choices. Observed BGP segments already encode ordered routes through multiple networks. We preserve this information by joining a selected prefix $q = (s, \ldots, z)$ to an intact forward segment $T_{z,j}(P)$ from Equation (1), including z once. The result must have 3–20 distinct valid ASes, admissible prefix edges, and a feasible valley-free completion of known relationships [6]. Unknown relationships remain unknown. These checks establish compatibility; the source's use of the constructed route still requires assessment.

To bound construction while retaining alternatives, we keep up to two evidence-prioritized candidates per first hop, actual third AS, destination, and AS count. Priority favors fewer inferred prefix extensions and support from more observation sources. A deterministic work budget covers 5,000 destination contexts, supplemented by protected contexts for sparse prefixes. These construction representatives are scored as complete routes.

Learning complete-path observation support. A LightGBM classifier $h_\theta(\phi(P))$ scores the 255-feature candidate representation, using 300 trees, at most 31 leaves per tree, depth seven, and learning rate 0.04. For a training source, a candidate is positive if its complete ordered sequence appears in that source's reference. A nonmatching sequence is negative only when its destination has reference coverage; uncovered destinations remain unknown and are excluded. Weighted binary cross-entropy balances sources, first hops, destinations within each first hop, and alternative paths per destination.

The scorer uses 160 training sources and 40 disjoint validation sources. Upstream predictions for a training source come from models fitted without its source fold; donor statistics exclude the same original-source fold even when recursion changes the stitching AS. This is stagewise source isolation, not fully nested end-to-end cross-fitting. Validation and target queries use the training models and donors. Final target reference paths do not determine features, selection, or checkpoints. Scores estimate finite-corpus observation support, not route-existence probabilities.

Preserving important forwarding branches. High scores alone can concentrate the library on a well-observed neighbor. We therefore combine support-based ranking with minimum sample requests for important neighbors. A neighbor is major if it is in the Top-3 of π_s or has $\pi_s(n) \geq 0.1$. For a nominal budget B clipped to candidate capacity, we request

$$L_s(n) = \begin{cases} \max\{50, \lceil 0.5\pi_s(n)B \rceil\}, & n \text{ is major,} \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

Requests are clipped to available paths and shortages are recorded. The budget can expand to retain the available mandatory paths. This preserves samples through important neighbors without requiring equal branch counts.

After these requests, selection greedily fills the remaining budget using the marginal gain of

$$F(\mathcal{R}) = \sum_{P \in \mathcal{R}} \log \max\{h_\theta(\phi(P)), \epsilon\} - \lambda \sum_d c_d(\mathcal{R}) \log c_d(\mathcal{R}), \tag{6}$$

where c_d counts selected paths to d, $0 \log 0 = 0$, $\epsilon = 10^{-12}$, and $\lambda = 0.1$. The weak repetition penalty discourages redundant destination sampling; selection uses a greedy approximation.

Integrating augmented and observed paths. Algorithm 1 applies this selection in two passes. First, generated candidates form an augmentation shortlist $\mathcal{U}_s$, with nominal size 1,000 and a goal of 1,000 paths absent from both the public path index and $\mathcal{K}_s^{\mathrm{tr}}$, the subset of full suffixes provided by training sources. We retain the highest-scoring overall and previously unavailable candidate per first-hop–destination–AS-count group. Branch and novelty requirements can expand the budget; insufficient candidate capacity is recorded. If $|\mathcal{U}_s| < 1{,}000$, we return it as $\mathcal{R}_s$ without a fusion pass.

Otherwise, we integrate eligible full suffixes from $\mathcal{K}_s$. Each preserves its observed direction and original terminal AS; its first hop must belong to $\mathcal{N}_s$ and occur in $\mathcal{U}_s$. The pool is

$$\mathcal{V}_s = \mathcal{U}_s \cup \mathcal{K}_s^{\mathrm{eligible}}. \tag{7}$$

We deduplicate ordered sequences, mark external suffixes and their overlaps as previously available, and score the pool with the same model and representative rule. A second selection uses budget $|\mathcal{U}_s|$

Algorithm 1 Evidence-guided Path Augmentation

Require: Source s, selected neighbors $\mathcal{N}_s$, importance π_s
Require: Observed segments $\mathcal{T}$, suffixes $\mathcal{K}_s$, $\mathcal{K}_s^{\text{tr}}$
Require: Next-hop predictor g, support scorer h_θ
Require: $\tau = 300$, $D = 3$, $k = 10$, $B_0 = \nu_0 = 1{,}000$
Ensure: Selected source-specific library $\mathcal{R}_s$

$\mathcal{B} \leftarrow \emptyset$; $W \leftarrow [(s, n) : n \in \mathcal{N}_s]$
while $W \neq \emptyset$ **do**
  $q \leftarrow \textsc{Pop}(W)$
  $c \leftarrow \textsc{UsableTailCount}(q, \mathcal{T}, \tau)$
  $E \leftarrow \{q \oplus T : T \in \mathcal{T}_{\text{last}(q)}, \textsc{Valid}(q, T)\}$
  $\mathcal{B} \leftarrow \mathcal{B} \cup E$
  if $c < \tau$ and $|q| < D$ **then**
    $A \leftarrow \textsc{SupportedTopK}(g(s, q), k)$
    for all $a \in A$ with $a \notin q$ **do**
      $\textsc{Push}(W, q \parallel (a))$
    end for
  end if
end while
$\mathcal{B} \leftarrow \textsc{ConstructionReps}(\mathcal{B})$
$\mathcal{B} \leftarrow \textsc{WorkBudget}(\mathcal{B}, 5000)$
$\textsc{MarkUnavailable}(\mathcal{B}, \mathcal{T}, \mathcal{K}_s^{\text{tr}})$
$\mathcal{B} \leftarrow \textsc{ScoreAndRepresent}(\mathcal{B}, h_\theta)$
$\mathcal{U}_s \leftarrow \textsc{Select}(\mathcal{B}, \pi_s, B_0, \nu_0)$
if $|\mathcal{U}_s| < B_0$ **then**
  return $\mathcal{U}_s$ ▷ record capacity shortfall
end if
$\mathcal{K}' \leftarrow \{P \in \mathcal{K}_s : \textsc{ValidFull}(P),$
  $\textsc{FirstHop}(P) \in \mathcal{N}_s \cap \textsc{FirstHops}(\mathcal{U}_s)\}$
$\mathcal{V} \leftarrow \textsc{Deduplicate}(\mathcal{U}_s \cup \mathcal{K}')$
$\textsc{MarkKnown}(\mathcal{V}, \mathcal{K}')$
$\mathcal{V} \leftarrow \textsc{ScoreAndRepresent}(\mathcal{V}, h_\theta)$
$\mathcal{R}_s \leftarrow \textsc{Select}(\mathcal{V}, \pi_s, |\mathcal{U}_s|, 0)$
return $\mathcal{R}_s$ for both training and candidate generation

and the same branch safeguards, without another novelty minimum. Thus alternatives to the same destination compete on support and branch needs. The final library $\mathcal{R}_s$ retains provenance for every selected path.

In Algorithm 1, $q \oplus T$ joins a prefix and segment at their common AS, and $q \parallel (a)$ appends a successor. $\textsc{Select}(X, \pi, B, \nu)$ applies the branch requests and greedy objective, with nominal budget B and minimum ν previously unavailable paths when capacity permits.

4.5 Web-Path Inference

Our implementation uses GMPI [19] for the score in Equation (2). Its training and local candidate generation consume the same selected library $\mathcal{R}_s$. Remaining public observations support topology-based candidate generation and history retrieval. GMPI keeps candidate and historical paths as separate inputs and ranks query candidates; the support scorer governs library admission. Queries return one estimated route, or no estimate if usable candidates are absent. Same-AS and direct-edge cases undergo the same reference-based evaluation.

5 Evaluation

5.1 Research Questions

We evaluate the tool in two complementary settings: a BGP experiment that withholds source-originated paths, and traceroute measurements toward domain-derived endpoints. The BGP setting represents destination ASes as potential Web-service networks. The domain-endpoint setting provides data-plane observations toward actual resolved destinations.

The evaluation asks three questions: **RQ1**, how accurately does the tool estimate routes from a user network to a serving network; **RQ2**, do selected first hops and anchors cover the observed routing choices; and **RQ3**, do constructed paths represent the selected neighbors and have complete-path observation support?

5.2 Data and Source Withholding

Table 3 lists data sources, dates, scales, and roles.

Controlled BGP setting. We use three monthly Route Views and RIPE RIS RIB snapshots: February 1, March 1, and April 1, 2026, at 08:00 UTC. For each date, ten sources were selected by the largest original per-source path files. All distinct source–destination pairs form the fixed query sets. The March snapshot included 52 Route Views and 23 RIS collectors. After removing prefixes and duplicate AS sequences, its construction graph contained 74,781 ASes and 537,872 edges. Its sources were 16552, 19529, 32590, 34927, 42473, 49544, 58057, 199524, 328840, and 398465.

Before constructing the graph or source-specific evidence, paths starting at these sources were withheld. Other sources' paths could still contain them internally and provide forward suffixes.

Domain-endpoint setting. We resolved the frozen Tranco list to IPv4 targets on October 15 and probed from AS63835 through October 16. We retained the earliest TCP observation per destination and used ICMP Paris traceroute for targets not reached by TCP. The 631,698 records are measurements, not independent AS-pair queries.

We mapped hop addresses using the dated CAIDA prefix-to-AS snapshot and collapsed consecutive copies of an ASN. Under the same-AS gap assumption, an unknown run bounded by that ASN was removed; gaps between different ASes or at an endpoint were not filled. Eligibility required reaching the target, a single destination-origin mapping, matching endpoints, no remaining unknown hop, and no AS loop. The resulting 1,890 AS pairs include one same-AS pair. No raw record had a fully mapped continuous TTL sequence without this assumption. The endpoints were domain-derived, not individually verified HTTP services.

All available October 15 BGP evidence was used; AS63835 had no originating path in that corpus. Traceroute paths were excluded from graph construction, augmentation inputs, training, and checkpoint selection. They supplied the reference observations for scoring the fixed queries.

5.3 Implementation Settings

We used the GMPI ranker with 256-dimensional AS representations and a hidden size of 512. The BGP benchmark includes KnownPath [25] and PathRadar [10] as reference methods. For the domain-endpoint experiment, we used batch size 64, Adam learning rate 0.001, and a maximum of 30 training epochs. A stable 10% split of construction-derived training pairs provided internal validation, with early stopping after three non-improving epochs. This experiment used random seed 20261015. Reference paths were excluded from model training and checkpoint selection. For the domain-endpoint queries, we compared with KnownPath [25], which extends known AS paths under routing constraints. Both methods were scored on the same query pairs.

Table 3: Data sources, snapshot dates (2026, month–day), and uses.

Data source	Snapshot / period	Contents, scale, and use
Route Views / RIPE RIS [29, 32]	02-01, 03-01, 04-01; 10-15	Public RIB paths supply the graph, segments, and training evidence. Ten withheld sources per month yield 685,820 / 685,765 / 688,302 test pairs; October BGP supplies evidence for Web queries.
CAIDA AS Relationships [3]	02-01, 03-01, 04-01; 10-01	Provider–customer and peer labels for neighbor features and routing-policy compatibility checks; missing relationships remain unknown.
CAIDA ASRank [5]	02-01, 03-01, 04-01; 10-01	AS and link attributes, including rank, degree, and customer-cone size, for neighbor selection and path scoring.
PeeringDB [24]	02-01, 03-01, 04-01; 10-14	Network attributes and IXP / facility memberships for candidate retrieval and interconnection features; shared presence does not establish adjacency.
Tranco N26XW [15]	10-14	Frozen list of 1,000,000 ranked domains for selecting Web measurement targets.
DNS resolution (this study)	10-15	911,257 resolved domains yield 388,491 unique IPv4 probe targets and domain-to-endpoint associations.
CAIDA prefix-to-AS [4]	10-13, 10:00 UTC	1,126,405 IPv4 prefix-to-origin entries for longest-prefix mapping of endpoints and hops; ambiguous origins remain ambiguous.
Traceroute (this study)	10-15–10-16	AS63835: 388,491 TCP + 243,207 ICMP-Paris records. Under the same-AS gap-removal rule, 22,844 eligible records yield 2,198 distinct ordered reference paths over 1,890 AS pairs, used only for evaluation.

The final Jaccard scores were calculated from returned path sequences against the reference observations, including queries handled by same-AS and direct-edge rules.

5.4 Metrics at Each Stage

Final query agreement. We retain AS-set Jaccard as the primary end-to-end metric. Let $S(P)$ be the set of ASes in route P, including endpoints, and let $\mathcal{T}(q)$ contain the distinct reference paths for fixed query q:

$$J(P,T) = \frac{|S(P) \cap S(T)|}{|S(P) \cup S(T)|}, \qquad \mathrm{Acc}_J = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{round}_3 \left(\max_{T \in \mathcal{T}(q)} J(\widehat{P}_q, T) \right). \tag{8}$$

Missing or endpoint-invalid predictions contribute zero to (8). Each fixed pair has equal weight, regardless of domain aliases or observation multiplicity. Jaccard measures AS-set agreement; it does not establish AS order, exact routes, or routing correctness for every prefix.

Neighbor and anchor selection. First-hop coverage C_1 is the fraction of reference routes whose first neighbor is selected, averaged within each pair and then across eligible pairs. Same-AS paths have no first hop and are excluded. Fixed-sparse anchor coverage C_2 uses sources of graph degree at most ten and reference paths with at least three ASes and a selected first hop. It checks whether the third AS is an admitted, enabled anchor. Selected branches with no anchor, exhausted budgets, or disabled recursion score zero. The Web diagnostic further conditions on branches flagged to attempt recursion, and is named trigger-conditional anchor coverage. Both average within a pair and then over eligible pairs. Their denominators are reported separately.

Branch representation. Let O_s be the forward first-hop neighbors witnessed in the reference paths, and $\mathcal{S}_s = \mathcal{N}_s \cap O_s$. For each selected-and-witnessed neighbor, we report whether any constructed path was generated and whether at least one generated path matched a complete ordered reference. The two coverage denominators are $\sum_s |\mathcal{S}_s|$; zero-output branches remain included. An admitted neighbor absent from the reference is marked unverified, rather than false. Equal generated counts are not required, and neither the observed nor generated distribution is interpreted as traffic volume.

Table 4: Path-estimation results across three monthly BGP snapshots. KnownPath uses 10,000 sampled pairs per date; PathRadar and WebRouteMap use the full query sets.

Date	Method	Scored pairs	Jaccard
2026-02-01	KnownPath	10,000	72.10%
	PathRadar	685,820	68.73%
	WebRouteMap	685,820	80.02%
2026-03-01	KnownPath	10,000	70.43%
	PathRadar	685,765	68.46%
	WebRouteMap	685,765	78.99%
2026-04-01	KnownPath	10,000	72.15%
	PathRadar	688,302	68.44%
	WebRouteMap	688,302	80.31%
Mean	KnownPath	10,000	71.56%
	PathRadar	686,629	68.54%
	WebRouteMap	686,629	79.77%

Complete-path observation support. Let $\mathcal{R}_Q$ contain distinct constructed paths whose endpoint pairs occur in Q. The support rate is

$$U = \frac{|\{P \in \mathcal{R}_Q : P \in \mathcal{T}(\mathrm{pair}(P))\}|}{|\mathcal{R}_Q|}. \tag{9}$$

Equation (9) requires complete ordered equality, not Jaccard equal to one. Paths outside Q lack a reference and are reported separately. We also recompute support after excluding constructed paths already present as full source-rooted sequences in $\mathcal{K}_s$. An unmatched path is unsupported by this finite reference, not proven nonexistent.

5.5 BGP Path-Visibility Results

RQ1: Returned path views. Table 4 summarizes path-estimation results across three monthly BGP snapshots. The stage-wise assessment below examines the 685,765 query pairs in the March snapshot.

RQ2: Exit and anchor coverage. Selected first hops covered 84.52% of the observed routes under pair-balanced aggregation. On the sparse-source cohort of 137,394 eligible pairs, anchor coverage C_2 was 85.10%. These metrics assess the inferred prefix separately from

Table 5: Selection and constructed-path assessment for the BGP setting. Branch coverage uses the 143 selected-and-witnessed neighbors. Complete-path support uses the 140,762 constructed paths with reference-covered AS pairs.

Metric	WebRouteMap
Admitted source–neighbor relations	152
Selected, reference-witnessed neighbors	143
Witnessed branches with generated paths	142/143
Witnessed branches with complete-path support	135/143
First-hop coverage C_1	84.52%
Sparse-source anchor coverage C_2	85.10%
Distinct constructed paths	141,277
Reference-evaluable constructed paths	140,762
Ordered supported paths	106,326
Complete-path support U	75.54%
Support excluding known full sequences	74.54%
Paths without a reference pair	515

Table 6: Path estimates toward domain-derived endpoints from AS63835. Both methods are scored on the same 1,890 AS pairs, with missing predictions contributing zero. Reference sequences use the same-AS gap-removal assumption.

Method	Valid outputs	Jaccard
KnownPath	1,866	69.19%
WebRouteMap	1,837	69.96%

the complete route: a covered exit does not establish that every continuation through that exit is used by the source.

RQ3: Branch representation and complete-path support. Table 5 reports the evidence-construction results. Of 152 admitted source–neighbor relations, 143 were witnessed in the reference paths. Constructed paths represented 142 of these branches (99.30%); 135 branches (94.41%) had at least one constructed path matching a complete ordered reference. The branch from AS328840 to AS328913 produced no admissible path. The method therefore retained evidence across most selected, witnessed neighbors, without guaranteeing coverage of every branch.

Among the 140,762 evaluable constructed paths, 106,326 matched a complete ordered reference sequence, giving 75.54% support. After excluding paths already available as known full source-rooted sequences, support was 74.54% (98,420 of 132,028 paths). This distinguishes support for assembled routes from reuse of known complete routes. The 515 paths without a reference pair were not assigned a support score.

5.6 Domain-Endpoint Results

The domain-endpoint experiment evaluated the network paths behind Web access from AS63835. Table 6 reports the same fixed-pair Jaccard metric for WebRouteMap and KnownPath.

WebRouteMap obtained 69.96% Jaccard, compared with 69.19% for KnownPath. KnownPath returned more valid outputs, while WebRouteMap had higher mean AS-set agreement over the full query set. This result concerns the single measured source and the selected endpoint mappings.

The selected first hop was AS4134, with $C_1 = 99.09\%$ over 1,889 cross-AS pairs. Recursive anchors AS1299, AS2914, and AS3356 gave trigger-conditional anchor coverage of 58.37% over 1,879 eligible pairs. This illustrates the single-neighbor setting: knowing the first hop leaves choices about the continuation beyond that neighbor.

The construction produced 12,849 distinct paths. Of these, 744 had a reference-covered AS pair, and 395 matched a complete ordered reference, giving 53.09% support on the evaluable subset. The remaining 12,105 paths lacked a reference pair and were not scored as either supported or false. Consequently, 53.09% describes the reference-covered subset, not all constructed paths.

6 Discussion

What the path view could support. A user could compare estimated routes to news, shopping, and video endpoints to identify recurring transit networks or inspect networks of concern. A content provider could issue the same queries from several user ASes toward candidate serving ASes and examine differences in estimated network dependencies. The tool supplies routing information for such analysis. It does not itself steer traffic around an AS, select a deployment, or measure the resulting performance or security benefit. AS-set agreement also does not validate an individual AS-avoidance decision.

Network and measurement scope. The ten BGP sources were selected for their large original path files. Withholding their paths does not make them representative of poorly observed access networks. The Web measurement adds one measurement vantage, not global user coverage. Endpoint mapping, prefix-specific policies, intra-AS routing, and routing drift limit the AS-pair view. The same-AS gap assumption can conceal an unobserved detour, and incomplete traceroute responses do not establish a packet-perfect route.

Evidence and validation limits. Compatibility checks and learned observation-support scores cannot prove that a constructed path is selected for a particular prefix. The BGP cohort also informed method development; its results do not establish generalization to unseen dates. The domain-endpoint experiment adds a complementary measurement setting, but uses one source and one random seed. Broader source and date coverage, repeated training, score calibration, and sensitivity to library selection remain limitations of the present evaluation.

7 Conclusion

We have presented WebRouteMap, a tool for estimating the networks behind Web access from shared public evidence. It constructs reusable source-specific paths through learned neighbor selection, evidence-capacity expansion, and complete-path scoring, then uses an existing ranker to answer user-network–endpoint queries. The three-snapshot BGP benchmark obtains approximately 80% AS-set Jaccard, and the single-source Web experiment obtains 69.96%. Stage-wise evaluation distinguishes selected exits, represented branches, and supported complete paths, exposing limitations hidden by a final aggregate score. The resulting view supports further study of Web routing dependencies; deployment and security benefits require direct evaluation.

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